Farseen T

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Summary

Full-Stack Developer and B.Tech graduate specializing in the MERN stack, with proven expertise in building secure, scalable web architectures integrating AI models and geospatial data. Skilled in JWT authentication, RBAC, and delivering high-performance, user-centric solutions.

Skills

Languages: JavaScript (ES6+), HTML5, CSS3, C, Java, Python

Frontend: React.js, Tailwind CSS, Leaflet.js, Responsive Design

Backend: Node.js, Express.js, Flask, RESTful APIs, JWT Authentication

Database: MongoDB

DevOps & CI/CD: Docker, Docker Compose, GitHub Actions, Vercel, Render

Security: Role-Based Access Control (RBAC), Authentication & Authorization

Tools: Git, GitHub, Postman, VS Code, Cloudinary

Projects

Academic Project

GeoSentinel | *MERN Stack* | *Sentinel Hub* | *Flask* | *Docker* *Oct 2025 – Mar 2026*

- Built a full-stack geospatial change detection platform where users draw an AOI on an interactive Leaflet map, select two dates, and submit satellite imagery analysis requests via the Copernicus Data Space (Sentinel Hub) API.
- Integrated Sentinel Hub in Node.js with OAuth authentication, BICUBIC upsampling, and dynamic resolution calculation; developed a Flask microservice interfacing with a FresUNet deep learning model to generate heatmaps, overlays, and edge maps.
- Implemented async background processing in Node.js/Express for non-blocking AI inference with real-time request status tracking and a React results dashboard with tabbed navigation and lightbox viewer.
- Containerized the full 4-service stack (React, Node.js, Flask, MongoDB) using Docker and Docker Compose with isolated networking, persistent volumes, and images published to Docker Hub enabling one-command deployment.

Academic Project

Transit-Hub | *MERN Stack* | *JWT* | *RBAC* *Jan 2025 – May 2025*

- Designed and developed a full-stack MERN application to improve communication between commuters and transport authorities, serving 500+ daily users.
- Implemented JWT-based authentication and RBAC for 4 distinct user roles; built RESTful APIs for buses, routes, reviews, complaints, and schedules.
- Created an Admin Dashboard with system analytics, audit logging, and full CRUD operations for buses, routes, and schedules.

Publications

AICTE Sponsored International Conference on Optimization Techniques for Learning (ICOTL'26)

An AI-Based Multi-Temporal Satellite Image Change Detection System for Environmental Monitoring *Mar 2026*

Farseen T - Dept. of Information Technology, IET University of Calicut

Experience

Bridgeon Solutions **Kakkanchery, Kozhikode**

Intern *Dec 2024*

- Developed and maintained 5+ web applications using HTML, CSS, JavaScript, and React, achieving 95%+ code coverage for core user-facing functionality.
- Led frontend development for a major e-commerce application using React and Tailwind CSS, implementing 10+ responsive pages; deployed on Vercel managing production builds and environment configuration.
- Collaborated with cross-functional teams in an Agile environment, participating in daily stand-ups and sprint planning to deliver features on schedule.

Education

University of Calicut **Thenhipalam, Kerala**

B.Tech in Information Technology *2022 – 2026*